Inference

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Outline

Logical inference
- Set theory and logic
- Logical inference

Probability background
- Probability facts
- Conditional probability and independence
- Posterior distributions and model estimation
- Random variables, expectation and variance

Graphical models
- Graphical model
- Exercises

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Set theory

- First, consider some universal set Ω.
- A set A is a collection of points x in Ω.
- $\{x \in \Omega : f(x)\}$: the set of points in Ω with the property that $f(x)$ is true.

Unary operators

- $\neg A = \{x \in \Omega : x \notin A\}$.

Binary operators

- $A \cup B$ if $\{x \in \Omega : x \in A \vee x \in B\}$ - (c.f. $A \vee B$)
- $A \cap B$ if $\{x \in \Omega : x \in A \wedge x \in B\}$ - (c.f. $A \wedge B$)

Binary relations

- $A \subset B$ if $x \in A \Rightarrow x \in B$ - (c.f. $A \implies B$)
- $A = B$ if $x \in A \Leftrightarrow x \in B$ - (c.f. $A \Leftrightarrow B$)

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The inference problem

- Given statements $A_1, \ldots, A_n$ we know to be true (i.e. a knowledge base), is another statement B true?

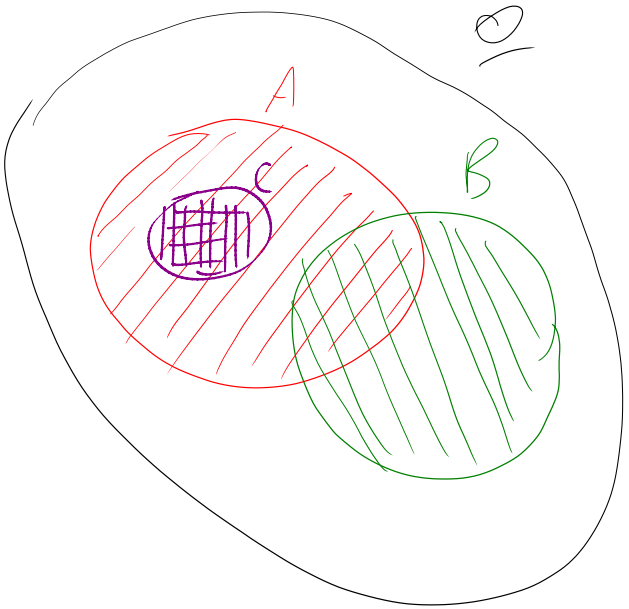
The following statements are equivalent:

- $A \implies B$ iff $(A \cap \neg B) = \varnothing$.
- $A \implies B$ iff $A \subset B$.

In addition

- If $(A \Rightarrow B) \wedge A$ then B.
- If $(A \wedge B)$ then A.

Illustration



$(A \mid C) =$

infered known

$(B \mid C) =$

$(C \mid A) =$

$(A \wedge B \mid C)$

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Events as sets

The universe and random outcomes

- The Ω contains all events that can happen.
- When something happens, we observe an element $\omega \in \Omega$.

Events in the universe

- An event is true if $\omega \in A$, and false if $\omega \notin A$.
- The negative event $\neg A = \Omega \setminus A$ is the set $\{\omega \in \Omega \wedge \omega \notin A$.
- The possible events are a collection of subsets Σ of Ω so that

(i) $\Omega \in \Sigma$, (ii) $A, B \in \Sigma \Rightarrow A \cup B \in \Sigma$ (iii) $A \in \Sigma \Rightarrow \neg A \in \Sigma$

Example: Traffic violation

- A car is moving with speed $\omega \in [0, \infty)$ in front of the speed camera.
- $A_0 = [0, 50]$: below the speed limit
- $A_1 = (50, 60]$: low fine
- $A_2 = (60, \infty]$: high fine
- $A_3 = (100, \infty)$: Suspension of license

All combinations of the above events are interesting.

Probability fundamentals

Probability measure P

Probability can be seen as an area-like function assigning a likelihood to sets.

- $P : \Sigma \to [0, 1]$ gives the likelihood $P(A)$ of an event $A \in \Sigma$.
- $P(\Omega) = 1$
- For $A, B \subset \Omega$, if $A \cap B = \varnothing$ then $P(A \cup B) = P(A) + P(B)$.

Marginalisation

Partition

If $A_1, \ldots, A_n$ are a partition of B then:

- $A_j \cap A_i = \varnothing$ for $i \neq j$
- $\bigcup_{i=1}^{n} A_i = B$.

Marginalisation

If $A_1, \ldots, A_n \subset \Omega$ are a partition of Ω

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i).$$

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Conditional probability

Definition (Conditional probability)

The conditional probability of an event A given an event B is defined as

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

The above definition requires $P(B)$ to exist and be positive.

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Conditional probabilities as a collection of probabilities

More generally, we can define conditional probabilities as simply a collection of probability distributions:

$$\{P_\theta : \theta \in \Theta\},$$

where Θ is indexing possible values of θ.

- θ is sometimes called the model or parameter
- P_θ is sometimes called the likelihood

Independence

Definition (Independence)

A, B are independent if

- $P(A|B) = P(A)$
- or equivalently $P(A \cap B) = P(A)P(B)$.

This means that knowing B does not tell us anything about A and vice-versa.

Definition (Conditional Independence)

A, B are conditionally independent given C if

- $P(A|B, C) = P(A|C)$
- or equivalently $P(A \cap B \mid C) = P(A|C)P(B|C)$.

This means that knowing B does not tell us anything about A and vice-versa.

Conditional probability game

You are each given a pair of cards, either black (♠, ♣) or red (♦, ♥).

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Card 1	Car 2	Prob.
R	R	1/3
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B	B	1/3

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If everybody opens one card, what is the proportion of cards that are going to be Red?

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Open	Closed	Pr.	$Pr(R \mid Open)$
R	R	2/6	$2/6 × 3/1
R	B	1/3	
B	R	1/3	
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Calculation from conditional probability

$P(R_2|R_1) = P(R_2 \cap R_1) \ / \ P(R_2)$

The theorem of Bayes

Theorem (Bayes's theorem)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \qquad P(B) = P(B|A)P(A) + P(B|\neg A)P(\neg A)$$

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Proof.

By definition of conditional probability:

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By marginalisation:

$$P(B) = P(B \cap A) + P(B \cap \neg A) = P(B|A)P(A) + P(B|\neg A)P(\neg A).$$

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Bayes's theorem: the general case

If $A_1, \ldots, A_n$ are a partition of Ω, meaning that they are mutually exclusive events (i.e. $A_i \cap A_j = \varnothing$ for $i \neq j$) such that one of them must be true (i.e. $\bigcup_{i=1}^{n} A_i = \Omega$), then

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$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

and

$$P(A_j|B) = \frac{P(B|A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

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Bayes's theorem

As a conditional probability

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)}$$

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As a causal explanation

$$\mathbb{P}(\text{cause} \mid \text{effect}) = \frac{\mathbb{P}(\text{effect} \mid \text{cause})\,\mathbb{P}(\text{cause})}{\mathbb{P}(\text{effect})}$$

As model inference

- Model family: $\{P_\theta : \theta \in \Theta\}$.
- Parameter: $\theta \in \Theta$.
- Prior: $\beta(\theta)$.
- Data: x, sampled from P_{θ^*}.
- Likelihood: $\{P_\theta(x) : \theta \in \Theta\}$ (in model family)
- Posterior:

$$\beta(\theta \mid x) = \frac{P_\theta(x)\beta(x)}{\sum_{\theta' \in \Theta} P_{\theta'}(x)\beta(\theta')}$$

Example: COVID symptoms

Activity (with playing cards or dice)

- Pick two (x, y) from 1 to 10.
- If ($x = 1$ and $y < 9$), or (x is even and $y \geq 9$), you have symptoms.
- Do you have COVID?

Example: COVID symptoms

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Information

- 20% of people have COVID
- 50% of people with COVID have symptoms.
- 10% of people with no COVID have symptoms.
- If you do have symptoms, what are the chances you have COVID?

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Formalisation

- Prior $P(C) = 0.1$:
- Likelihood: $P(S|C) = 0.5$, $P(S|\neg C) = 0.1$
- Posterior:

$$P(C|S) = \frac{P(S|C)P(C)}{P(S|C)P(C) + P(S|\neg C)P(\neg C)}$$

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Random variables

A random variable $f : \Omega \to \mathbb{R}$ is a real-valued function, with $\omega \sim P$.

[1] Formally, for A in a σ-algebra.

Random variables

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Definition (Uniform distribution)

P is a uniform distribution on $\Omega \iff P(A) = |A|/|\Omega| \ \forall A \subset \Omega$.[1]

Bernoulli distribution

If $f(\omega) = \begin{cases} 1, & \omega \leq \theta \\ 0, & \omega \geq \theta \end{cases}$, with P uniform on Ω, then we say that f is Bernoulli distributed with parameter θ and write

$$f \mid \theta \sim \mathrm{Bernoulli}(\theta)$$

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Exponential distribution

If \$f$(\omega) = \frac{\ln(\omega)}{\lambda}$ and ω is uniform

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Distributions of random variables

The distribution of f

The probability that f lies in some subset $A \subset \mathbb{R}$ is

$$P_f(A) \triangleq P(\{\omega \in \Omega : f(\omega) \in A\}),$$

and we write $f \sim P_f$.

Distributions of random variables

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The density of f

When f takes an infinite number of values, it is easier to talk about the density p_f, which is linked with the probability through integration:

$$P_f(A) \triangleq \int_A p_f(x)\, dx.$$

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Shorthands for RV

- For RVs $f : \Omega \to \mathbb{R}$, we write $P(f \in A)$ to mean $P_f(A)$.
- For RVs $f : \Omega \to X$, where X is a finite set e.g. $\{1, 2, \ldots, n\}$, we write $P(f = x) = P_f(\{x\})$ for any $x \in X$, or even $P(f)$.
- For continuous RVs $f : \Omega \to \mathbb{R}$, we use densities as $p_f(x)$.

Independence of random variables

Two RVs f, g are independent in the same way that events are independent.

$$P(f \in A \wedge g \in B) = P(f \in A)P(g \in B) = P_f(A)P_g(B).$$

In that sense, $f \sim P_f$ and $g \sim P_g$.

Formal definition

More specifically, we are measuring the set of ω values for which $f(\omega) \in A$ and $g(\omega) \in B$:

$$P(\{\omega : f(\omega) \in A, g(\omega) \in B\}) = P_f(A)P_g(B).$$

Shorthand notation

Since the above is very cumbersome, we usually just write that

$$P(f, g) = P(f)P(g)$$

for any two independent random variables f, g.

Expectation

Discrete probability space Ω

For any real-valued random variable $f : \Omega \to \mathbb{R}$, the expectation with respect to a probability measure P is

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Continuous probability space

When Ω is continuous, we define the probability $P(A)$ of any subset $A \subset \Omega$ through a probability density p:

$$P(A) = \int_A p(\omega) d\omega.$$

We can then define the expectation through the density:

$$\mathbb{E}_P(f) = \int_\Omega f(\omega) p(\omega) d\omega.$$

Properties of expectations

The law of the unconscious statistician

If $x : \Omega \to X$, with $X \subset \mathbb{R}$, and $\omega \sim P$, we can say that $k \sim P_x$, with $P_f(A) = P(\{\omega : f(\omega) \in A\})$.

$$\mathbb{E}_P(f) = \sum_{\omega \in \Omega} f(\omega) P(\omega) = \sum_{k \in X} k P_f(k).$$

Linearity of expectations

For any RVs x, y:

$$\mathbb{E}_P(x + y) = \mathbb{E}_P(x) + \mathbb{E}_P(y)$$

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We give the proof for the discrete case:

$$\begin{aligned}\mathbb{E}_P(x + y) &= \sum_{\omega \in \Omega} [x(\omega) + y(\omega)]P(\omega) \\ &= \sum_{\omega \in \Omega} x(\omega)P(\omega) + \sum_{\omega \in \Omega} y(\omega)P(\omega) = \mathbb{E}_P(x) + \mathbb{E}_P(y)\end{aligned}$$

Distributions of multiple variables

The joint distribution $P(x, y)$

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For two (or more) RVs $x : \Omega \to \mathbb{R}$, and $y : \Omega \to \mathbb{R}$, this is a shorthand for the distribution of $(x(\omega), y(\omega))$ when $\omega \sim P$. We can also use $P(x = i, y = j)$ for the probability that the two variables assume the values i, j respectively.

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Independence

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If x, y are independent RVs then $P(x, y) = P_x(x)P_y(y)$.

Correlation

If x, y are not correlated then $\mathbb{E}_P(xy) = \mathbb{E}(x)\,\mathbb{E}(y)$.

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IID (Independent and Identically Distributed) random variables

A sequence x_t of r.v.s is IID if $x_t \sim P$ so that

$$(x_1, \ldots, x_t, \ldots, x_T) \sim P^T$$

i.e. a T-length sample is drawn from the product distribution $P^T = P \times P \times \cdots \times P$.

Conditional expectation

Conditional expectations are similar to conditional probabilities.

Discrete Ω

The conditional expectation of a random variable $f : \Omega \to \mathbb{R}$, with respect to a probability measure P conditioned on some event B is simply

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$$\mathbb{E}_p(f|B) = \int_{\Omega} f(\omega) p(\omega|B) d\omega.$$

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$$P(x, y) = P(x|y)P(y).$$

This equation works for all possible values of x, y e.g.

$$P(x = 1, y = 0) = P(x = 1|y = 0)P(y = 0)$$

which then denotes the probability mass of each

Probability notation: math versus statistics

- $P(C)$: Probability of event C
- $P(A|B)$: Probability of the event A if we know B
- $P(A \cap B)$: Probability of A and B
- $P(A \cup B)$: Probability of A or B

- $P(x)$: distribution of variable x.
- $P(x|y)$: distribution of x for different values of y
- $P(x, y)$: joint distribution of x, y
- No correspondence.

Example: The k-meteorologists problem (set notation)

- R_t: The event that it rains at time t.
- A set of stations Θ, with $\theta \in \Theta$ making weather predictions:

$$P(R_{t+1} \mid R_1, \ldots, R_t, \theta),$$

- A prior probability $P(\theta)$ on the stations.
- The marginal probability

$$P(R_1 \cap \cdots \cap R_t) = \sum_{\theta \in \Theta} P(R_1 \cap \cdots \cap R_t \mid \theta) P(\theta)$$

- The posterior probability

$$P(\theta \mid R_1 \cap \cdots \cap R_t) = \frac{P(R_1 \cap \cdots \cap R_t \mid \theta) P(\theta)}{P(R_1 \cap \cdots \cap R_t)} = \frac{\prod_{i=1}^{t} P(R_t \mid R_1 \cap \cdots \cap R_{t-1}}{P(R_1 \cap \cdots \cap R_t)}$$
$$= \frac{P(R_t \mid R_1 \cap \cdots \cap R_{t-1} \mid \theta) P(\theta \mid R_1 \cap \cdots \cap R_{t-1})}{P(R_t \mid R_1 \cap \cdots \cap R_{t-1})}$$

- The marginal posterior probability

$$P(R_{t+1} \mid R_1 \cap \cdots \cap R_t) = \sum_{\theta \in \Theta} P(R_{t+1} \mid R_1 \cap \cdots \cap R_t, \theta) P(\theta \mid R_1 \cap \cdots \cap R_t)$$

Example: The k-meteorologists problem (stat notation)

- $x_t \in \{0, 1\}$: A random variable, telling us whether it rains at time t.
- A set of stations Θ, with $\theta \in \Theta$ making weather predictions:

$$P_\theta(x_{t+1} \mid x_1, \ldots, x_t)$$

- A prior probability $\beta(\theta)$ on the stations.
- The marginal probability

$$\mathbb{P}_\beta(x_1, \ldots, x_t) = \sum_{\theta \in \Theta} P_\theta(x_1, \ldots, x_t)\beta(\theta)$$

- The posterior probability

$$\beta(\theta \mid x_1, \ldots, x_t) = \frac{P_\theta(x_1, \ldots, x_t)\beta(\theta)}{\mathbb{P}_\beta(x_1, \ldots, x_t)} = \frac{\prod_{i=1}^{t} P_\theta(x_t \mid x_1, \ldots, x_{t-1})\beta(\theta)}{\mathbb{P}_\beta(x_1, \ldots, x_t)}$$
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- The marginal posterior probability

$$\mathbb{P}_\beta(x_{t+1} \mid x_1, \ldots, x_t) = \sum_{\theta \in \Theta} P_\theta(x_{t+1} \mid x_1, \ldots, x_t)\beta(\theta \mid x_1, \ldots, x_t)$$

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Graphical models

- A graphical model or Bayesian network is used to model dependencies between random variables.
- Each node in the graph is a variable.
- The arcs show what variable is an input to which variable.

Independence

Independent events $A \perp\!\!\!\perp B$

- A, B are independent iff $P(A \cap B) = P(A)P(B)$.
- Knowing if A happened, does not tell us anything about whether B happened

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- A, B are conditionally independent given C iff $P(A \cap B|C) = P(A|C)P(B|C)$.
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For random variables

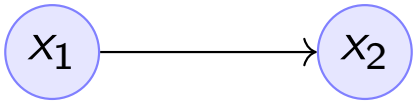
- Independence: $P(x, y) = P(x)P(y)$.
- Conditional independence: $P(x, y|z) = P(x|z)P(y|z)$.

Model specification: Independent

$$f = \text{Bernoulli}(1/2)$$
$$g = \text{Bernoulli}(0.8)$$
$$x_1 \sim f$$
$$x_2 \sim g$$

```
def f():
  return np.random.choice(2)
def g:
  return np.random.choice(2, [0.2, 0.8])
x1 = f()
x2 = g()
```

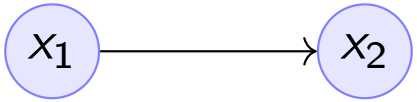
Model specification: Gaussian Dependent variables



$$f = \mathrm{Normal}(0, 1)$$
$$g(a) = \mathrm{Normal}(a, 1)$$
$$x_1 \sim f$$
$$x_2 | x_1 = a \sim g(a)$$

```
def f():
  return np.random.normal(0,1)
def g(a):
  return np.random.normal(a)
x1 = f()
x2 = g(x1)
```

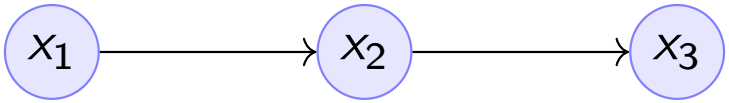
Model specification: Bernoulli Dependent variables



$$
\begin{aligned}
f &= \text{Bernoulli}(1/2) \\
g(a) &= \text{Bernoulli}(\theta_a) \\
x_1 &\sim f \\
x_2 | x_1 = a &\sim g(a) \\
\theta &= (0.6, 0.5)
\end{aligned}
$$

```
def f():
  return np.random.choice(2)
def g(a):
  theta = [0.6, 0.5]
  return np.random.choice(2,
 [1 - theta[a], theta[a]])
x1 = f()
x2 = g(x1)
```

Model specification: Chain



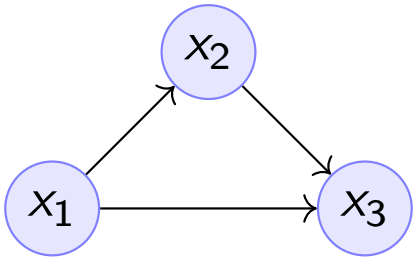
$$x_1 \sim f \tag{1}$$

$$x_2 \mid x_1 = a \sim g(a) \tag{2}$$

$$x_3 \mid x_2 = b \sim h(b), \tag{3}$$

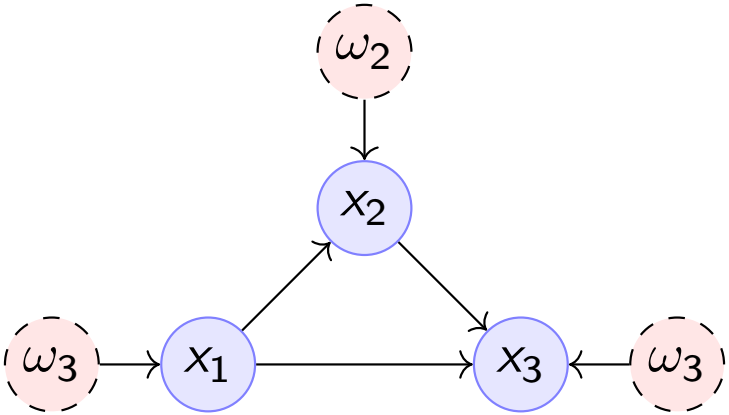
```
def f():
  return np.random.uniform()
def g(a):
  return np.random.uniform() + a
def h(b)
  return np.random.uniform * b
x1 = f()
x2 = g(x1)
x3 = h(x2)
```

Graphical models



- Variables: x_1, x_2, x_3
- Arrows denote dependencies between variables.
- In this example the value of x_3 is a function of x_1, x_2, as well as a random input.

Graphical models



- Variables: x_1, x_2, x_3
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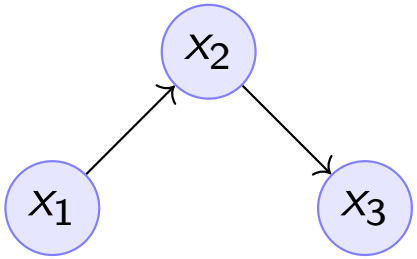
- In this example the value of x_3 is a function of x_1, x_2, as well as a random input.

Conditional independence

Example



Graphical model for the factorisation $\mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$.

Definition

- Consider variables $x_1, \ldots, x_n$.
- Let B, D be subsets of $[n]$.

We say x_i is conditionally independent of $\boldsymbol{x}_B$ given $\boldsymbol{x}_D$ and write

$$x_i \perp\!\!\!\perp \boldsymbol{x}_B \mid \boldsymbol{x}_D$$

if and only if:

$$\mathbb{P}(x_i, \boldsymbol{x}_B \mid \boldsymbol{x}_D) = \mathbb{P}(x_i \mid \boldsymbol{x}_D)\,\mathbb{P}(\boldsymbol{x}_B \mid \boldsymbol{x}_D).$$

Conditional independence

We can write the joint distribution of any set of RVs $x_1, \ldots, x_n$, as

$$P(x_1, \ldots, x_n) = \prod_i P(x_i \mid x_1, \ldots, x_{i-1})$$

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$$P(x_1, \ldots, x_n) = \prod_i P(x_i \mid x_1, \ldots, x_{i-1})$$

However, we can use a Bayesian network to define conditional independence structures.

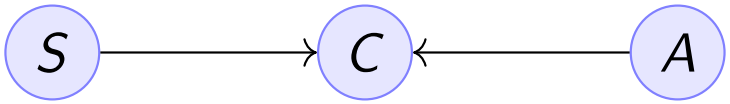
Bayesian Networks and Independence

If A is a parent of B and C is a child of B, and there are no other paths from A to C then the following conditional independence holds:

$$P(C \mid B, A) = P(C \mid B)$$

i.e. C is conditionally independent of A given B.
This holds more generally for sets of random variables.

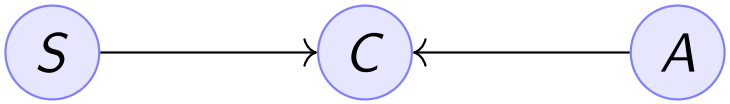
Smoking and lung cancer



Smoking and lung cancer graphical model, where S: Smoking, C: cancer, A: asbestos exposure.

- Here, S, A are independent
- However, they become dependent if we know C.

Smoking and lung cancer



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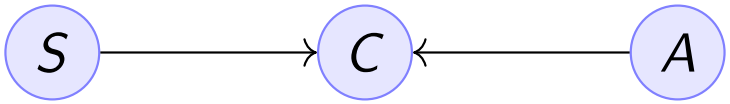
$$P(S, C, A) = P(S)P(A)P(C|S, A)$$

Smoking and lung cancer



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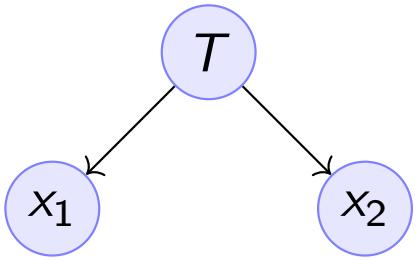
- Here, S, A are independent
- However, they become dependent if we know C.

$$P(S, C, A) = P(S)P(A)P(C|S, A)$$

$$\begin{aligned} P(A, S|C) &= P(A|S, C)P(S|C) \\ &= \frac{P(C|A, S)P(A|S)}{P(C|S)} \frac{P(C|S)P(S)}{P(C)} \\ &= \frac{P(C|A, S)P(A|S)}{P(S|C)P(C)/P(S)} \frac{P(C|S)P(S)}{P(C)} \end{aligned}$$

Time of arrival at work

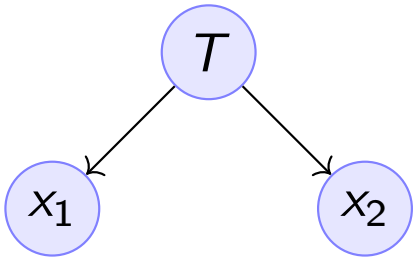


Time of arrival at work graphical model where T is a traffic jam and x_1 is the time John arrives at the office and x_2 is the time Jane arrives at the office.

Conditional independence:

- Even though x_1, x_2 are not independent, they become independent once you know T.

Time of arrival at work



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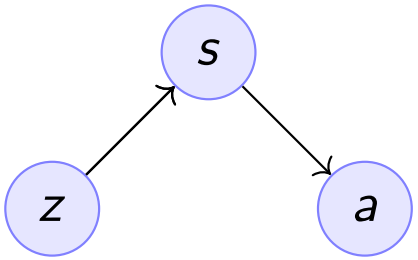
$$
\begin{aligned}
P(x_1, x_2, T) &= P(x_1, x_2 | T) P(T) \\
&= P(x_1 | T) P(x_2 | T) P(T) \\
P(x_1, x_2 | T) &= P(x_1 | T) P(x_2 | T) P(T) \\
P(x_1, x_2) &= \sum_{i=0}^{1} P(x_1 | T = i) P(x_2 | T = i) P(T = i)
\end{aligned}
$$

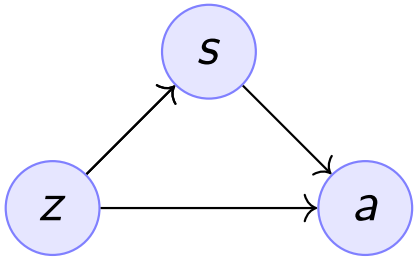
School admission

School	Male	Female
A	62	82
B	63	68
C	37	34
D	33	35
E	28	24
F	6	7

- z: gender
- s: school applied to
- a: admission



Is admission independent of gender?



How about here?

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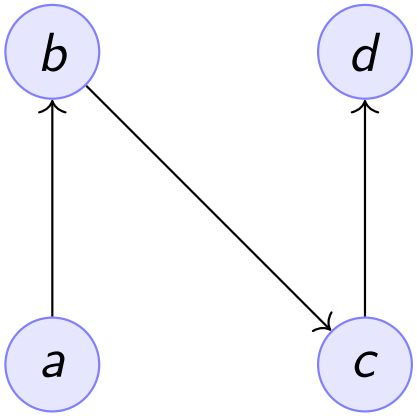
What is the model for this graph?



$$P(a, b, c, d) = \cdots$$

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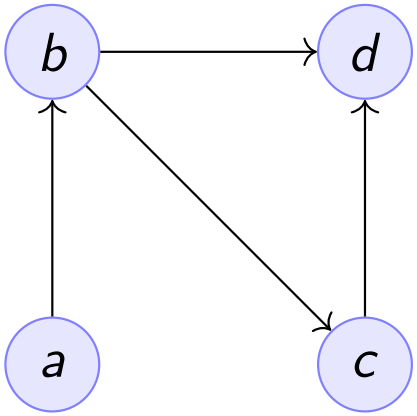

$$P(a, b, c, d) =$$

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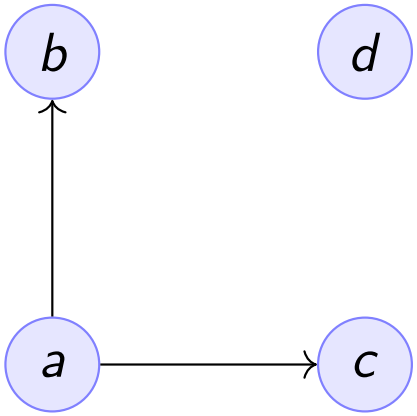
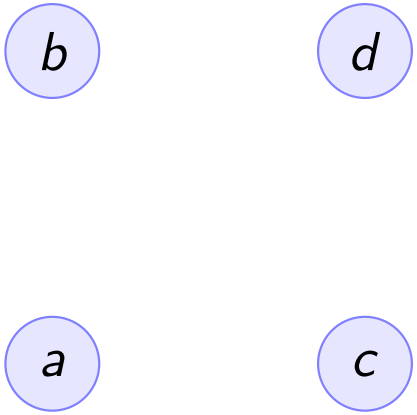


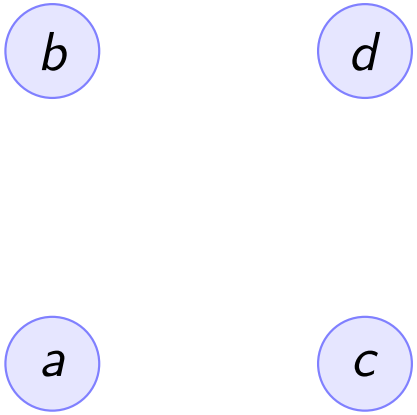
$$P(a, b, c, d) =$$

Draw the graph for this model



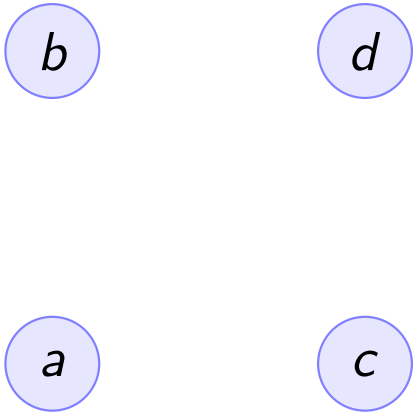
$$P(a, b, c, d) = P(a)P(b|a)P(c|b)P(d|b)$$

Draw the graph for this model



$$P(a, b, c, d) = P(a)P(b|a)P(d|c)P(c)$$

Draw the graph for this model



$$P(a, b, c, d) = P(a)P(b|a)P(c|a)P(d|b, c)$$

Example: COVID test

Information

- 10% of people have COVID
- 50% of people with COVID have a positive test
- 50% of people with COVID have symptoms
- 10% of people without COVID have a positive test
- 20% of people without COVID have symptoms

Example: COVID test

Information

- 10% of people have COVID
- 50% of people with COVID have a positive test
- 50% of people with COVID have symptoms
- 10% of people without COVID have a positive test
- 20% of people without COVID have symptoms

Formalisation

- Prior: $P(C = 1) = 0.1$
- Likelihood: $P(T, S|C) = P(T|C)P(S|C)$, $P(T, S|\neg C)$ for all values of T, S, C.
- Posterior:

$$P(C|T, S) = \frac{P(S|C)P(T|C)P(C)}{\sum_{i=0}^{1} P(S|C = i)P(T|C = i)P(C = i)}$$

Example: Naive Bayes models

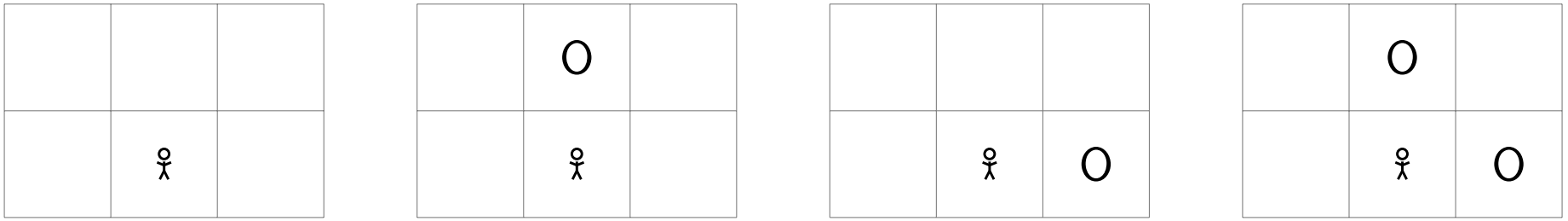
Sometimes we observe multiple effects that have a common cause, but which are otherwise independent:

$$\mathbb{P}(\text{effect}_1, \ldots \text{effect}_n \mid \text{cause}) = \prod_{i=1}^{n} \mathbb{P}(\text{effect}_i \mid \text{cause})$$

Naive Bayes model

- Observations $(\boldsymbol{x}_t, y_t)_{t=1}^{T}$ with $\boldsymbol{x}_t = (x_{t,1}, \ldots, x_{t,n})$.
- Probability models $P_\theta(y \mid \boldsymbol{x}) = \prod_{i=1}^{n} P_\theta(y \mid x_i)$.

Example: Wumpus world



Details

- Probability of each world A_i being true: $1/4$
- Probability of each hole generating a breeze: $P(B_1|A_2 \cup A_4) = P(B_2|A_3 \cup A_4)$ with B_1, B_2 conditionally independent given A.

Questions

- What is the probability of feeling a breeze $B = B_1 \cup B_2$ in each world?
- What is the probability of a hole above if you feel a breeze?
- What is the probability of a hole above f you don't feel a breeze?